

Markov Decision Processes DSEFM PhD course, University of Luxembourg

2026

Homework Exercises - Set 1

Infinite horizon criteria

Due date: Monday May 4th 2026 at 23:30 hours via email to joachim.arts@uni.lu or directly in moodle.

The homework set should be done individually but you are allowed/encouraged consult each other in your work. Submit a report as well as code for implementations of algorithms in any language that you like (Matlab, Python, R, Java, C, Fortran...)

Please define unambiguously all the variables you use and show your work clearly.

Question

For this assignment you will find the optimal policy and cost-rate/discounted cost for a MDP. You are free to choose a MDP that fits well with your research interest. For example, you can take a MDP from a paper that you like and re-implement. If there is no specific MDP that you fancy, I also provide one description of a problem below as well as references to some papers with MDPs in the Supply Chain Management application domain. If in doubt about which MDP to implement, please reach out to me and we can look into it together.

Below is a list of possible MDPs (which is not exhaustive; you may choose a MDP that is not on this list)

- (1) Consider an item that is subject to degradation according to a Poisson process with rate λ . This item is inspected every time unit and if the degradation exceeds the failure threshold τ , then it must be replaced at cost C_u . In any degradation state $x \in \{0, 1, \dots, \tau - 1\}$ it is possible to not replace the item, which does not incur any cost or to replace at a cost $C_p < C_u$. Your objective is to minimize the average cost of keeping the item up an running. This model is based on Kolesar (1966).
- (2) The lost sales inventory system is described in detail by Zipkin (2008). Fix a lead-time (my suggestion is either 1 or 2 time-units) and implement this MDP.
- (3) The production of blood platelets for healthcare can be controlled by formulating an appropriate MDP. Haijema et al. (2007) provides such an MDP. This MDP can only be solved for small problem instances.

- (4) Traffic control at an intersection can be modeled as an MDP. Haijema and van der Wal (2008) provide a good example. Also in this context it is important to only look at small problem instances.

Choose a MDP of your liking, either related to your own research or from the list above. For this MDP do the following:

- (a) Give a description of the state-space, the action-space, the immediate rewards (costs) and the transition probabilities.
- (b) If you look at the average cost criterion, argue why your MDP is unichain.
- (c) If you look at the average cost criterion, argue why your MDP is aperiodic. If it is not aperiodic, apply a transformation to make it such.
- (d) Solve the MDP numerically by value iteration.
- (e) Solve the MDP numerically by either policy iteration or linear programming. Verify that you obtain the same optimal cost-rates/discounted cost as with value iteration.
- (f) Conduct a small numerical investigation on how the optimal cost and/or optimal policy behaves as a function of one of the problem parameters (e.g. cost factors, lead times, arrival/demand intensity etc.)

References

- Haijema, R. and van der Wal, J. (2008). A MDP decomposition approach for traffic control at isolated signalized intersections. *Probability in the Engineering and Informational Sciences*, 22(4):587-602.
- Haijema, R., van der Wal, J., and van Dijk, N. M. (2007). Blood platelet production: Optimization by dynamic programming and simulation. *Computers & Operations Research*, 34(3):760 – 779. Logistics of Health Care Management.
- Kolesar, P. (1966). Minimum cost replacement under Markovian deterioration. *Management Science*, 12:694–706.
- Zipkin, P. (2008). Old and new methods for lost-sales inventory systems. *Operations Research*, 56(5):1256–1263.